

6.1.20. Let $B_i = \{x \in \mathbb{R} \mid 0 \leq x \leq i\}$ for each integer $i = 1, 2, 3, 4$.

- a. $B_1 \cup B_2 \cup B_3 \cup B_4 = ?$
- b. $B_1 \cap B_2 \cap B_3 \cap B_4 = ?$
- c. Are B_1, B_2, B_3, B_4 mutually disjoint?
Explain.

a. $B_1 \cup B_2 \cup B_3 \cup B_4 = ?$

We must show the union of B_1, B_2, B_3 and B_4 .

Definition:

$$A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$$

Let $B_i = \{x \in \mathbb{R} \mid 0 \leq x \leq i\}$

$$B_1 = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

$$B_2 = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$$

$$B_3 = \{x \in \mathbb{R} \mid 0 \leq x \leq 3\}$$

$$B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 4\}$$

$$B_1 \cup B_2 \cup B_3 \cup B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 1 \text{ or } 0 \leq x \leq 2 \text{ or } 0 \leq x \leq 3 \text{ or } 0 \leq x \leq 4\}$$

$$\text{Therefore, } B_1 \cup B_2 \cup B_3 \cup B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 4\}$$

Solution: $B_1 \cup B_2 \cup B_3 \cup B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 4\}$

b. $B_1 \cap B_2 \cap B_3 \cap B_4 = ?$

We must show the intersection of B_1, B_2, B_3 and B_4 .

Definition:

$$A \cap B = \{x \in U \mid x \in A \text{ and } x \in B\}$$

Let $B_i = \{x \in \mathbb{R} \mid 0 \leq x \leq i\}$

$$B_1 = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

$$B_2 = \{x \in \mathbb{R} \mid 0 \leq x \leq 2\}$$

$$B_3 = \{x \in \mathbb{R} \mid 0 \leq x \leq 3\}$$

$$B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 4\}$$

$$B_1 \cap B_2 \cap B_3 \cap B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 1 \text{ and } 0 \leq x \leq 2 \text{ and } 0 \leq x \leq 3 \text{ and } 0 \leq x \leq 4\}$$

$$\text{Therefore, } B_1 \cap B_2 \cap B_3 \cap B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

Solution: $B_1 \cap B_2 \cap B_3 \cap B_4 = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$

- c. Are B_1, B_2, B_3, B_4 mutually disjoint?
Explain.

We must show whether B_1, B_2, B_3, B_4 are mutually disjoint.

Definition:

$$A_i \cap A_j = \emptyset \text{ whenever } i \neq j$$

So by definition, we must show whether the sets have no elements in common.

But we know from solution (a) that $B_1 \subseteq B_2 \subseteq B_3 \subseteq B_4$.

Hence, they are not mutually disjoint, because B_4 contains elements from all other sets.

Solution: B_1, B_2, B_3, B_4 are not mutually disjoint.